

Multi-Agent System for Rogue Drone Interception

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Abstract—A multitude of applications based on unmanned aircraft vehicles (UAVs), in conjunction with the UAVs' diminishing costs and increasing capabilities, present a major threat to public safety as well as critical infrastructure security, leading to the need for versatile and robust counter-drone solutions. This work presents a novel multi-agent jamming system, where a group of cooperative autonomous agents employ various algorithms (detection, tracking, jamming, and self-localization) to counter unauthorized drone operations. The proposed system employs a joint wireless jamming and cooperative positioning framework to best track and intercept the moving rogue drone, utilizing a relative positioning system based on signals of opportunity in conjunction with inertial and vision measurements. Software-defined radio technology is incorporated on the pursuer agent to achieve rogue drone GPS disruption, while at the same time the autonomous agents cooperate to compute the location estimate of the pursuer agent. In the presence of jamming, to improve the wireless communication performance of the autonomous agents, a cellular communication architecture is also used as an additional channel for control and information exchange. For evaluation purposes, a prototype multi-agent system is designed, implemented, and tested in a real-world environment to demonstrate its enhanced localization and jamming performance, when compared to single-agent approaches.

Index Terms—Aerial systems: perception and autonomy, localization, multi-robot systems, sensor fusion.

I. INTRODUCTION

THE rapid technological advancements and exponential growth, in global demand, of UAVs, have led to a plethora of civilian and military applications. However, their ease-of-use introduces a vast array of threats and risks concerning security, public safety, and privacy [1]. Illegal/unauthorized drone operations over sensitive, restricted areas and critical infrastructures introduces the necessity of robust and accurate counter-drone systems. Unfortunately, to date, interception technologies are immature, with only a few advanced systems developed that are able to disrupt rogue drones (RDs) over restricted areas, utilized only for military purposes, and with extremely high price points. The most common countermeasure proposed for civilian applications is the utilization of general RF jamming

systems that are able to interfere with the drone's GNSS or WiFi reception signals [2]. However, the levels of interference induced by these jamming systems raise serious concerns regarding the normal operation of critical infrastructure systems [3].

Currently, a wide range of counter-drone systems are under investigation, with some of these systems utilizing software-defined radio (SDR)-based jamming and spoofing techniques, mainly due to the significant advancements in SDRs. Within such systems, SDR technology can also be employed to provide reliable and precise localization approaches, which is the primary requirement in autonomous vehicle navigation [4], principally by employing the fusion of various sensor modalities (such as visual, inertial, and optical flow (OF) measurements) along with signals of opportunity (SOPs) in GNSS-denied areas, including TV, AM/FM, and cellular signals [4], [5], [6], [7], [8]. In addition, several works have investigated the usage of a group of cooperative agents [9], [10], [11] to provide additional capabilities (such as reliability and robustness), as compared to single-agent solutions. However, no works have adequately focused on developing efficient multi-agent autonomous counter-drone systems that are also able to provide self-localization.

In accordance, this work proposes a novel multi-agent jamming system where a group of collaborating agents use SOPs and other sensor modalities for online distributed relative positioning for the pursuer agent, and also detection, tracking, and jamming of the rogue drone within an area of interest. Extending our previous autonomous single-agent counter-drone system as detailed in [12], the proposed system acts in a cooperative manner to continuously jam the rogue drone while simultaneously ensuring that, in GNSS-denied scenarios, each agent distributes its location in conjunction with vision data in order to provide relative localization for the pursuer agent. Specifically, the contributions of this work are as follows:

- An online multi-agent jamming system (MUAV-JS) prototype is implemented, both in terms of hardware and software, based on a group of cooperative autonomous agents. The designed system incorporates detection, tracking, jamming, and cooperative relative positioning technologies to fulfill the objectives of online rogue-drone track and follow, continuous jamming, and cooperative (robust) relative localization of the pursuer drone (PD) (by fusing SOP information with visual and inertial data) in case its own GPS is disrupted due to the jamming procedure.
- A communication architecture is employed as an additional channel allowing for robust control and information exchange between the agents, even in beyond visual line-of-sight operations.
- The implemented prototype is extensively tested in numerous outdoor experiments to assess its performance in terms of tracking-and-jamming the rogue drone, as well as self-localizing the pursuer drone, validating its feasibility.

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II. RELATED WORK

Several works in the literature focus on the design of practical counter-drone systems [13]. A number of these techniques rely on SDR; for example, in [14], a portable ground-based UAV GPS interference method is proposed, that uses GPS deception to change the UAV's location, as well as remote control communication signal suppression. Moreover, in [15], SDR technology is investigated in an outdoor environment by deploying different jamming methods, while in [16], an evaluation performance analysis is conducted with the use of an SDR-based GPS jamming approach. Further, [17] examines navigation interception in numerous real-life scenarios by employing a low-cost SDR platform for UAV GPS jamming.

In addition, a number of other (non-SDR-based) solutions have also been investigated for countering drone operations. In particular, [18] uses an RF-based mechanism that transmits high-power jamming signals to obstruct a rogue drone's communication link. Also, in [19], a counter-drone system is proposed for surveillance, localization, and frequency jamming of a rogue drone. However, the proposed system exhibits high energy consumption, wireless communication interference, and increased computational requirements.

In terms of localization when GNSS signals are unavailable, numerous attempts that incorporate SOPs have been investigated [4], [5], [6], as well as approaches that use inertial measurements and vision/optical flow data [11], [20]. In addition, in our work in [21], a relative positioning system (RPS) is proposed that fuses SOPs with inertial and optical flow modalities.

More recent works utilize autonomous single- or multi-agent counter-drone approaches, including the work in [12], where a prototype autonomous drone is able to achieve the two-fold objective of rogue drone jamming and self-localization based on signals of opportunity in conjunction with inertial measurements in real-life practical scenarios. Different techniques have been proposed that aim to achieve interception of rogue UAVs using UAV-based defense systems. Recently, the authors in [22], [23], proposed efficient and accurate multi-agent tracking-and-jamming schemes for rogue drones; nevertheless, the applicability and effectiveness of their approaches is only validated using simulations, with no practical implementation presented due to hardware and software complexity issues. Further, in [24], a counter UAV system (C-UAS) is presented that intercepts the malicious UAV utilizing a swarm defensive formation, while in [25], a cooperative group of quadrotors are positioned on the perimeter of a no-fly zone to quickly capture the evader once it emerges from it. An algorithm that coordinates the pursuer drones around the zone is presented, minimizing the capture time with the use of robust predictive control. Moreover, a distributed algorithm for cooperative pursuit of multiple evaders in a bounded environment is investigated in [26], where Voronoi tessellation is employed to capture all evaders in finite time, and with the technique evaluated in different environments. Finally, a swarm framework for autonomous detection of rogue UAVs in a defined airspace is examined in [27], providing a method for impeding the operation of rogue drones and detailing preliminary prototype software and hardware requirements.

This work complements the aforementioned research efforts by exploiting the benefits of a group of autonomous drones communicating in a distributed manner. The proposed system, with interception, detection, tracking, and localization functionalities, aims to counter one or more rogue drones in the area of interest. The novelty of the proposed system lies in the

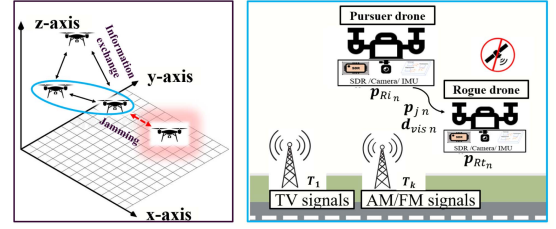


Fig. 1. System model.

multi-agent scheme's joint localization and jamming functionality introduced for efficiently tracking the rogue drone, while simultaneously navigating, based on cooperative localization, and continuously jamming the target to prevent it from completing its mission. Such a distributed scheme that combines all these processes together and implemented in a prototype, has not been previously investigated in the literature. Finally, contrary to other counter-drone techniques, the end goal of the MUAV-JS system is to intercept the rogue drone, thus allowing law enforcement to investigate the equipment and collect data in an effort to identify the malicious party. Of course, this is possible only in cases where the rogue drone is not designed so as to fly in a random direction or self-destruct upon recognizing that it has been detected and is under pursuit.

III. SYSTEM MODEL

The proposed system model (Fig. 1) assumes a group of UAVs (a total of $|M|$ UAV agents) that is launched to counter unauthorized operations over critical infrastructures. Each UAV employs various sensor modalities, based on its designated mission, to execute different functionalities such as relative self-localization, detection, tracking, or jamming in a 2D space. The UAV system is comprised of a selected pursuer agent (agent i), which is responsible for the target's detection-and-tracking and the jamming procedure at each time-step n , and a set of J UAV agents that collect and fuse sensor information in order to estimate and distribute the location of the pursuer agent. Specifically, each agent $j \in J$ is able to self-localize in cases where the GNSS information is unavailable utilizing its onboard SDR that collects SOPs, which are fused with inertial and visual measurement data. The position of all agents in J is subsequently combined to compute the pursuer agent's location, as a means to improve the dynamic tracking and jamming performance of the system. It should be noted, that each agent employs a cellular network communication architecture that enables the exchange of information between all agents, including the distribution of the pursuer agent's estimated location.

For the proposed system the following assumptions are made: (i) GNSS may become unavailable, (ii) a synchronous system is considered, (iii) agents follow a dynamic path, and (iv) the pursuer agent's position is estimated based on the information collected from agents $j \in J$, as the pursuer agent is not able to self-localize, with each agent able to communicate and exchange information with any other agent in the system.

The selection of the pursuer (agent i) is based on the Euclidean distance d_c between the target and the system's agents in M , which specifies that a target with state $\mathbf{p}_{Rt}$ at time n is detected by an agent with state $\mathbf{p}_{Ri} = [p_{Ri}(1) \ p_{Ri}(2)]^T$. Upon the selection of the i -th agent (pursuer), the $j \in J$ remaining

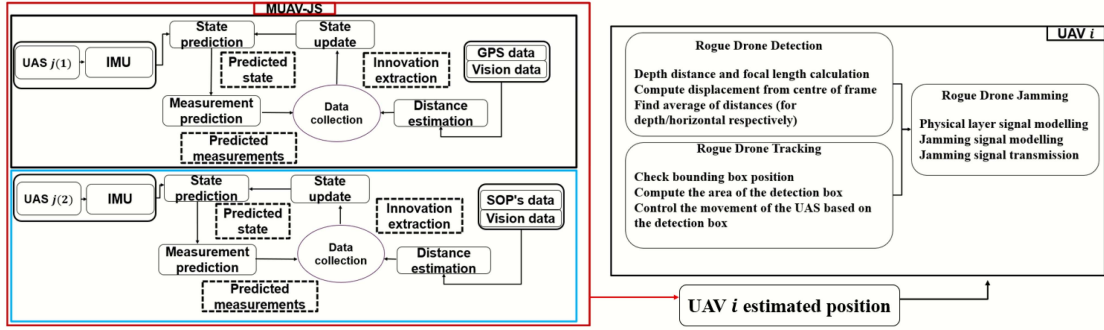


Fig. 2. MUAV-JS architecture overview.

agents are able to detect-and-track the pursuer agent, while also obtaining relative positioning $\mathbf{p}_{RPSj}$ by employing the SOP transmitters ($T_k, k \in \mathcal{K}$) in the area (with the use of an SDR), and fusing various sensor measurement from an onboard camera and the inertial measurement unit (IMU) (i.e., fusing online visual information, odometry, and SOPs for relative position estimation, navigating in a local relative frame). As the locations of the SOP transmitters, $\mathbf{p}_{T_k}$, are initially unknown, a non-linear least squares problem is solved, based on the assumption that in the initialization phase a GNSS fix is obtained, in order to estimate their locations [28]. At the same time, the i -th agent employs its camera to detect-and-track the target, and when the rogue drone is in range ($d_c \leq \bar{d}$), based on a pre-defined detection range $\bar{d}$, the jamming procedure is enabled.

As the pursuer agent cannot self-localize, due to its continuous use of the SDR for jamming, the remaining $j \in J$ agents are cooperating to provide its estimated location. Specifically, each agent j self-localizes via the use of SOPs, IMU, and vision data, leading to the generation of a number of relative trajectories. A detection-and-tracking algorithm is then employed to follow the pursuer agent. The Euclidean distances between the pursuer agent and agents $j \in J$ are computed and the pursuer agent's relative location is extracted. Thus, the proposed system is distributed, as each agent operates in its local position frame with its current starting position as the initial location, and it executes all algorithms autonomously, without requiring any data input from other agents. The MUAV-JS architecture is shown in Fig. 2.

IV. RELATIVE POSITIONING SYSTEM

A summary of the RPS methodology utilized by the autonomous agents $j \in J$ of the proposed MUAV-JS system is presented below. A detailed description of RPS can be found in our previous works in [21], [28], [29].

- **SOPs Model:** The received signal strength (RSS) data from various fixed SOP transmitters ($T_k, k \in \mathcal{K}$) are collected to compute vehicle position estimation ($\mathbf{p}_{R_j} = [p(1) \ p(2)]^T \in \mathbb{R}^2$). The SOPs follow a discretized model $\mathbf{p}_{T_k}^{n+1} = \mathbf{F}\mathbf{p}_{T_k}^n + \mathbf{w}$, where $\mathbf{F}$ and $\mathbf{w}$ are the system dynamics and process noise, respectively.
- **Multilateration & LSQ Techniques:** Using the path-loss (P-L) model, the distance $d(k, j)$ between the k -th ($k \in \mathcal{K}$) estimated SOP transmitter, ($\mathbf{p}_{T_k} \in \mathbb{R}^2$), and the location of the SDR receiver ($\mathbf{p}_{R_j}, j \in J$) mounted on the UAV is computed [30]. The estimated position is calculated using a multilateration and a least squares (LSQ) technique [31]

to solve an overdetermined system. Then, an approximated solution $\mathbf{p}^*$ (estimated receiver location) is obtained after the linearization of the system.

- **EKF Localization:** Finally, an extended Kalman filter (EKF) is used to extract the position estimates ($\hat{\mathbf{p}}$) of each agent by fusing different sensor measurements. In particular, the IMU data along with the SOP information are combined to calculate the relative position of the UAV.

V. DETECTION, TRACKING, AND JAMMING

In this work, a vision-based algorithm is used to select the pursuer drone (employing the Euclidean distance extracted from the vision data), to detect-and-track the rogue drone by agent i , and to detect-and-track agent i by agents $j \in J$.

- **Detection:** A large UAV sighting dataset is acquired and applied for the training phase (utilizing 10,000 labeled images). The detection module is established based on the Tiny YOLOv4 [32] convolutional neural network. For online application and high-accuracy performance, the object detection algorithm is applied on an onboard unit that processes the video frames received by the UAV's camera.
- **Tracking:** The tracking algorithm employed uses the intersection-of-union scores method and the Hungarian algorithm [33]. Specifically, the area of the detection box, computed at the detection step, is used to compute the movement of the pursuer drone in relation to the target. When the pursuer drone is within a certain distance d_c of the target, as detailed in our work in [29], the jamming procedure initiates. Hence, the effective jamming range is set so that the detected bounding box is between 7 – 10% of the image.
- **Jamming:** With the selection of the i -th pursuer agent, if the *sensing requirement* is met (i.e., $d_c \leq \bar{d}$), then the jamming procedure commences that targets the physical layer of wireless systems to disrupt the UAV's vital subsystems, such as its GPS. In this work, the barrage jamming method is utilized to transmit a rectangular pulse in the receiver's band in order to interfere with the rogue drone's received GPS signal, $y_{GPS}(t)$ [16], [17]. Specifically, prior to jamming, the received GPS signal is given by $y_{GPS}(t) = x_{GPS}(t) + n_{GPS}(t)$ ($x_{GPS}(t)$ is the transmitted GPS signal and $n_{GPS}(t)$ is the additive Gaussian noise at time t), while when jamming is present it becomes $y_{GPS}(t) = x_{GPS}(t) + x_{jam}(t) + n_{GPS}(t) + n_{jam}(t)$ ($x_{jam}(t)$ is the jamming signal and $n_{jam}(t)$ is the

jammer-rogue drone communication link additive Gaussian noise). It should be noted that the barrage jamming technique is employed, as it requires low energy and no knowledge of the target's own communication signals [17].

VI. MUAV-JS ALGORITHM

The proposed MUAV-JS algorithm is detailed below (Algorithm. 1). In general, the MUAV-JS framework is divided into the self-localization phase, related to each agent's relative self-localization (RPS) (for all autonomous agents $j \in J$), the distribution phase, where the estimated relative positions of agents $j \in J$ are utilized to compute the relative position of the i -th agent, and the jamming phase, where agent i continuously tracks-and-jams the target drone. At first, a brief explanation of the relative self-localization of each agent j is detailed, followed by a description of the MUAV-JS technique.

The relative positioning system (RPS) of each drone initiates with the acquisition of a number of $\mathbf{X}^n$ frequency sweeps, N_s , exploiting a set of frequency bands l , and the mean ψ_l^n (for each frequency band) is calculated as $\psi_l^n = \frac{1}{N_s} \sum_{n=1}^N \mathbf{X}_l^n, \forall l$. Afterwards, ψ_l^n is employed in the various localization methods (i.e., multilateration, LSQ, and discrete EKF) as detailed in [28] and summarized in Section IV, to compute the relative coordinates and extract position estimates ($\hat{\mathbf{p}}$) of each agent. Specifically, the relative position of each agent is predicted using the following state dynamics with sampling time T_s and with state information $\mathbf{p}^n = [p_R(1) \ p_R(2)]^T$ defined as the agent's planar position. The motion and measurement models are then:

$$\hat{\mathbf{p}}^{n+1} = \hat{\mathbf{p}}^n + T_s \begin{bmatrix} \cos \varphi^n & 0 \\ 0 & \sin \varphi^n \end{bmatrix} \left(\begin{bmatrix} u_x^n \\ u_y^n \end{bmatrix} + \mathbf{w}^n \right), \quad (1)$$

$$\hat{\mathbf{d}}^{n+1} = \sqrt{(x_{land} - \hat{p}_R(1))^2 + (y_{land} - \hat{p}_R(2))^2} + \nu^n, \quad (2)$$

with $\hat{\mathbf{p}}^{n+1}$ and $\hat{\mathbf{d}}^{n+1}$ denoting the position and distance measurement estimates at time-step $n+1$, respectively.

Then, for both the position and measurement estimates, denoted as functions $\hat{\mathbf{p}}^{n+1} = f(\hat{\mathbf{p}}^n, \mathbf{u}^n)$ and $\hat{\mathbf{d}}^{n+1} = h(\hat{\mathbf{p}}^n)$, respectively, the state dynamics and observation matrices are updated as follows:

$$\mathbf{F} = \left. \frac{\partial f}{\partial \hat{\mathbf{p}}^{n+1}} \right|_{\hat{\mathbf{p}}^n, \mathbf{u}^n}, \quad \mathbf{H} = \left. \frac{\partial h}{\partial \hat{\mathbf{p}}^{n+1}} \right|_{\hat{\mathbf{p}}^n}. \quad (3)$$

In this work, the process noise, $\mathbf{w}^n$, is modeled by a normal distribution with a zero mean and process noise covariance $\mathbf{Q} = (0.1)\mathbf{I}_{2 \times 2}$. Thus, the system's covariance is given by:

$$\hat{\mathbf{P}}^{n+1} = \mathbf{F}\hat{\mathbf{P}}^n\mathbf{F}^\top + \mathbf{Q}. \quad (4)$$

Then, the EKF prediction step follows, with the agent's heading angle and velocity measurements, φ^n and u_x/u_y , used as input to the motion model to generate the agent's state at time-step n . For the EKF update step, the measurement model is incorporated to compute the range between the various route points, with landmarks (i.e., x_{land}, y_{land} from each drone's SOP measurements in cases of GNSS unavailability) and ν^n the zero mean noise measurement with constant covariance $\mathbf{R} = 0.01$. Afterwards, the relative trajectory is generated, using the computed range measurements $\hat{\mathbf{d}}^{n+1}$ calculated employing the optical flow data as described in [21]. The Kalman gain $\mathbf{K}$

Algorithm 1: MUAV-JS Algorithm.

Input: Acquired RSS data at each flight route point ($\mathbf{X}^n$)

- 1: Sweep $\mathbf{X}^n$ at UAV current location and obtain $\mathcal{K}$
- 2: **while** $n < N$ (i.e., for a set number of steps) **do**
- 3: **MUAV-JS Procedure**
- 4: **procedure** (Detection)
- 5: Continuous video stream acquisition
- 6: x_{pixel}, y_{pixel} calculation
- 7: Detection and tracking of RD and PD
- 8: Compute depth distance and focal length
- 9: $\mathbf{f} = \frac{w_{RP} * d_r}{w_{R^m}}$ (d_r is the reference distance and w_{R^m} and w_{RP} the reference object width, in meters and pixels, respectively)
- 10: Compute Euclidean distance as $d_c = \frac{w_{R^m} * \mathbf{f}}{w_{RP}}$
- 11: Average for depth/horizontal distances calculated
- 12: x-y displacement distances in meters calculated
- 13: **end procedure**
- 14: **procedure** (Tracking)
- 15: Acquire detection box if RD detected
- 16: If detection box area computed $>$ max threshold area
- 17: PD moves back to avoid collision
- 18: elseif detection box area computed $<$ min threshold area
- 19: PD moves forward and continues to track
- 20: else stop the drone movement
- 21: If x displacement $<$ min x displacement threshold
- 22: The PD moves counter-clockwise
- 23: elseif x displacement $>$ max x displacement threshold
- 24: The PD moves clockwise
- 25: If y displacement $<$ min y displacement threshold
- 26: The PD moves upwards
- 27: If y displacement $>$ max y displacement threshold
- 28: The PD moves downwards
- 29: else stop the drone movement
- 30: **end procedure**
- 31: Distribution moments ψ_m^n extracted using $\mathbf{X}^n$
- 32: Use ψ_m^n to compute relative distances via P-L model
- 33: Multilateration and LSQ applied
- 34: IMU data fusion with SOPs
- 35: Relative position $\hat{\mathbf{p}}^{n+1}$ is estimated using EKF
- 36: **procedure** (Jamming)
- 37: The pursuer agent is selected based on d_c
- 38: Detection and tracking of the RD
- 39: If $d_c \leq \bar{d}$
- 40: Jamming procedure is enabled
- 41: Detection and tracking of the pursuer agent
- 42: Exchange distances between agent i and all agents $j \in J$
- 43: **end procedure**
- 44: **end while**

Output: Cooperative relative traject. & RD interception

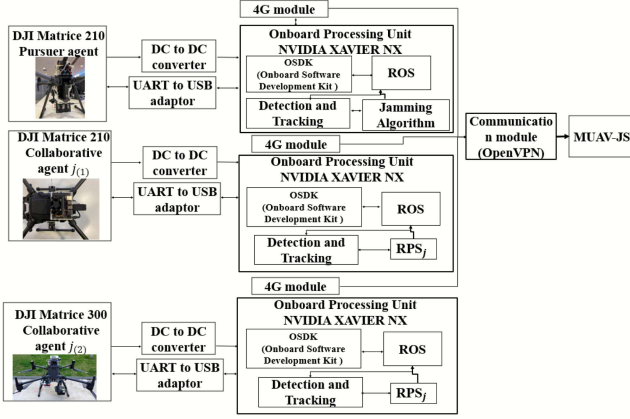


Fig. 3. Hardware implementation of the system.

is determined as $\mathbf{K} = \hat{\mathbf{P}}^{n+1} \mathbf{H}^\top (\mathbf{H} \hat{\mathbf{P}}^{n+1} \mathbf{H}^\top + \mathbf{R})^{-1}$, and the relative coordinates along with the error covariance at time-step $n + 1$ are computed as:

$$\hat{\mathbf{p}}^{n+1} = \hat{\mathbf{p}}^n + \mathbf{K}(\mathbf{d}^{n+1} - h(\hat{\mathbf{p}}^n)), \quad (5)$$

$$\hat{\mathbf{P}}^{n+1} = (\mathbf{I} - \mathbf{K}\mathbf{H})\hat{\mathbf{P}}^n. \quad (6)$$

When the rogue drone is in range, the agent with the smallest computed Euclidean distance (d_c) from the target is selected as the pursuer agent (agent i). The jamming phase of the rogue drone initiates when the sensing requirement is met (i.e., $d_c \leq \bar{d}$). Concurrently, the $j \in J$ autonomous agents initiate a cooperative relative position methodology (via the exchange of information regarding their position estimates), by using the vision data, and the relative position estimates from all $j \in J$ agents of the proposed system, in order to compute and update the pursuer agent's position, while at the same time the pursuer agent continuously jams the rogue drone. Let the position estimates extracted from all agents $j \in J$ be:

$$\hat{\mathbf{p}}_{MUAV-JSi}^{n+1} = \hat{\mathbf{p}}_{RPSj}^{n+1} + \mathbf{d}_{vis} \quad \forall j \in J, \quad (7)$$

where $\mathbf{d}_{vis}$ denotes the x-y displacement extracted from the vision measurements of each agent j . These estimates are subsequently utilized to generate the relative position estimate of the pursuer agent:

$$\bar{\mathbf{p}}_{MUAV-JSi}^{n+1} = \mathbb{E} [\hat{\mathbf{p}}_{MUAV-JSi}^{n+1}]. \quad (8)$$

Finally, in the MUAV-JS system, if the pursuer agent's energy level falls below a pre-defined threshold (15% in this case) and the rogue drone is still not disabled, a new pursuer agent is selected (from the group of J agents) based on each agent's Euclidean distance from the target.

VII. PERFORMANCE EVALUATION

A. Hardware and Software Set-up

The proposed MUAV-JS implementation consists of three UAV agents (see Fig. 3). Each autonomous drone agent is equipped with an onboard processing unit, the NVIDIA Jetson Xavier NX Developer Kit, that runs online all the algorithms and processes the different data modalities. Also, a 4 G dongle is utilized for the transmission of data and commands, while an SDR (HackRF One, with specifications as detailed at <https://>

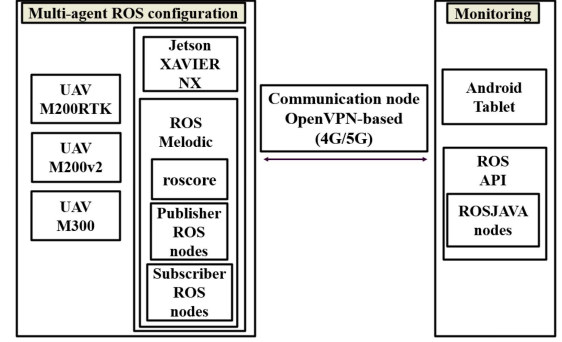


Fig. 4. ROS multi-master implementation of the system.

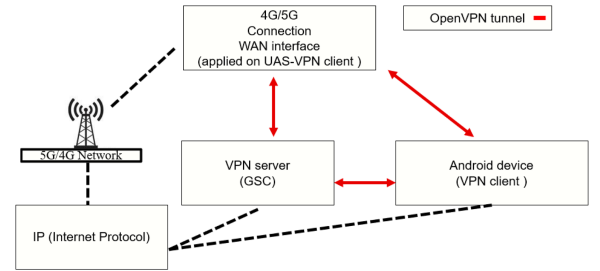


Fig. 5. Communication architecture.

greatscottgadgets.com/hackrf/one/) is applied on each agent for the dynamic sensing of SOPs. For the software implementation, ROS v1 is employed as the middleware to enable multi-agent interaction and distribute the various data modalities [34] (see Fig. 4).

B. Communication Architecture

In this work, to control, monitor, and exchange information between the autonomous agents, a 4 G/5 G communication architecture is implemented, as shown in Fig. 5. The robot operating system (ROS) framework is utilized to exchange data, and the communication module on an onboard processing unit is used to ensure online response on the agent. The proposed communication architecture consists of a VPN server (OpenVPN in this case), enabling bidirectional communication between UAV agents via a client-to-client protocol.

C. MUAV-JS Experimentation

This section showcases the MUAV-JS experimental results by comparing the localization performance for each trajectory waypoint with the ground truth (i.e., measurements acquired via GPS+IMU), and also demonstrates the vision, communication, and jamming performance evaluation. For the outdoor experiments conducted, the RPS configuration parameters employed are $|T_k| = 13$ (the number of SOP transmitters), $n_{PL} = 2.8$ (the path loss exponent value), and 0 – 3500 MHz (the SDR frequency range), based on our analysis in [28]. Further, the proposed multi-agent system is compared to our previous work in [12] which demonstrated a single agent counter-drone system with jamming and self-localization capabilities, revealing, nevertheless, a jamming duration and positioning accuracy tradeoff, due to the use of a single SDR that switches between the two aforementioned procedures. It is important to note that, in this work, the fusion of GPS+IMU is considered as the ground

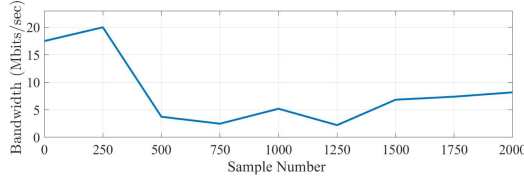


Fig. 6. MUAV-JS bandwidth in real-outdoor environment.

truth (GT). Both GPS and IMU have inherent measurement uncertainties associated with the system and electronic circuitry employed. Hence, in order to obtain the uncertainty concerning the GT, each experiment was repeated with the same parameters for the system setup and all measured GPS+IMU values were collected and analyzed. Thereafter, the proposed MUAV-JS's jamming procedure was enabled, thus the pursuer as well as rogue drone were not able to obtain accurate GPS measurements. In this way, a comparison was made with the GT values (i.e., GPS+IMU) collected when the experiments were repeated with the same parameters but without the jamming module enabled.

It should be also noted that the UAV agents are fitted with omnidirectional antennas; the selected pursuer agent, at each time-step n , transmits a barrage jamming signal of an average power spectral density level $P = -47$ dBW/Hz that is applied at the L1 band (GPS band at 1575 MHz that can be re-configured due to SDR hardware) to disrupt the rogue drone's autonomous or manual operation. Specifically, the rogue drone is continuously tracked and jammed until it is disabled. To investigate the average performance of the proposed system, 40 real outdoor experiments were conducted in the area of the University of Cyprus (UCY) campus. In these experiments, the rogue drone follows numerous pre-defined paths (i.e., straight, gamma, as well as more complex routes) over a range of velocities (5 – 12 m/s). The assumption in this work is that the rogue drone does not perform any evasive maneuvers.

First, a performance analysis concerning the communication architecture is presented in Fig. 6, illustrating that the communication of the proposed multi-agent system can be maintained continuously even at an altitude of 500 m and Euclidean distance from the remote controller of 2 km.

Further, the vision-based algorithms (i.e., detection and tracking) are evaluated, with Fig. 7(a) demonstrating that the MUAV-JS agents are able to detect-and-track a drone up to a Euclidean distance of 93 m. Specifically, a neural network is trained, with performance metrics related to training, validation and testing as depicted in Fig. 7(b), to detect a rogue drone in images, and then, the relative change in the position of the visual target based on an observer is computed and projected on a 2D space. To extract the optical flow-based range and displacement measurements, a dense OF algorithm is applied that utilizes each video frame captured by the onboard camera. Specifically, the Farneback OF algorithm is employed that can measure the next pixel position between the two given frames in a pre-defined region of interest, mainly due to its accuracy in online applications [35]. In particular, at first the video frame information is gathered online and a gray-scale conversion is applied to each frame, which is subsequently employed as input information to the Farneback OF algorithm. This algorithm, at each time-step n , uses as a reference point the initial video frame and compares it with the video frames that follow [36]. Our experimental results demonstrated that, with the use of optical flow measurements, the tracking performance increases from 80% to 94%.

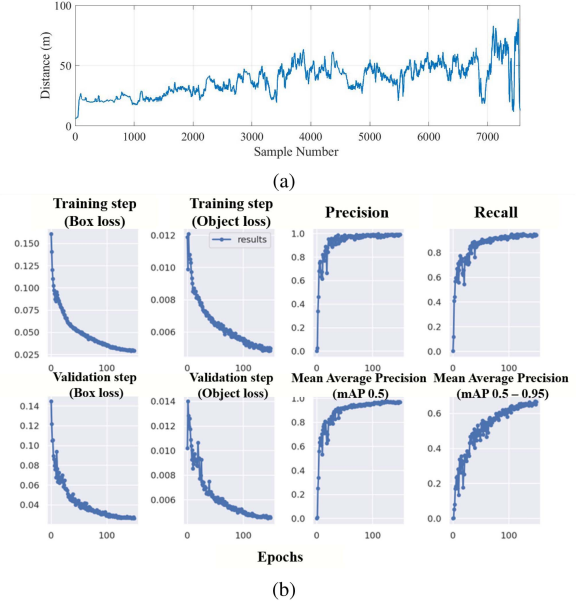


Fig. 7. (a) Tracking performance over distance (in meters) (b) Detector performance metrics.

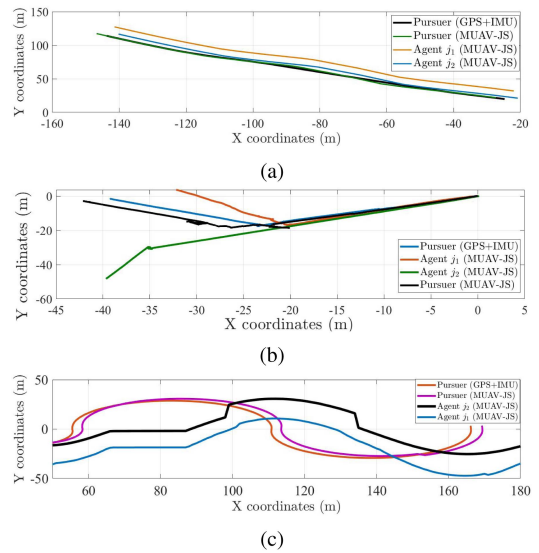


Fig. 8. MUAV-JS relative trajectories compared to GPS+IMU following the various maneuvers of the rogue drone.

The relative paths extracted for the UAV agents using the MUAV-JS framework are presented in Fig. 8. These results demonstrate that the pursuer drone is able to continuously follow a rogue drone, as well as navigate in the area by employing the relative trajectory (as estimated by the rest of the agents and based on the rogue drone's path). As shown, the MUAV-JS system, that uses various sensor modalities (i.e., SOPs in conjunction with IMU, optical flow, and vision data), can accurately estimate the pursuer agent's route, computing a relative trajectory that is analogous to the ground truth (i.e., the GPS+IMU measurements). Thus, by employing the proposed distributed localization method, online robust localization of the pursuer drone can be achieved.

Apart from the relative trajectories extracted for each agent, the difference between real and relative paths obtained by the

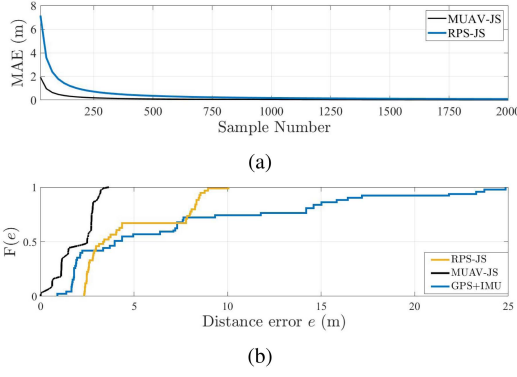


Fig. 9. (a) MAE (in meters) and (b) Position estimation error CDF of MUAV-JS and RPS-JS vs. GPS+IMU.

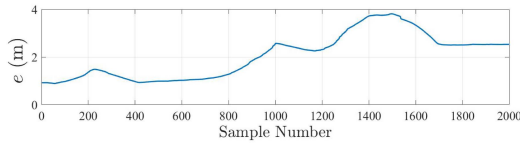


Fig. 10. Euclidean distance (in meters) of MUAV-JS vs. GPS+IMU.

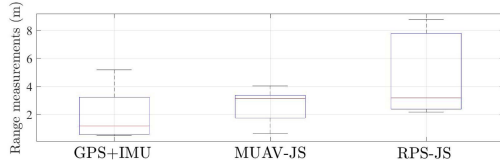


Fig. 11. Statistical comparison of GT, MUAV-JS, and RPS-JS. Range measurements are defined as the UAV distances covered from one waypoint to another.

MUAV-JS (multi-agent) and RPS-JS (single-agent) techniques, are shown in Fig. 9. The increase in accuracy of MUAV-JS compared to RPS-JS is evident in Fig. 9(a), as the mean absolute error (MAE) of MUAV-JS is lower compared to RPS-JS. Clearly, the employment of the RPS data from agents $j \in J$ compensates for the lack of self-localization for the pursuer agent. This is also shown in Fig. 9(b), that uses a cumulative distribution function (CDF) plot to illustrate that the localization error (Euclidean distance) of MUAV-JS reaches up to 4 m as compared to GPS+IMU. On the contrary, the RPS-JS error reaches values up to 10 m due to the localization and jamming procedure tradeoff (switching between the two procedures).

In addition, Fig. 10 demonstrates the deviation (in meters) of the relative trajectories for MUAV-JS in comparison to both the RPS-JS [12] and the ground truth (fused GPS+IMU). Also, as illustrated in Fig. 11, MUAV-JS provides higher localization accuracy as compared to RPS-JS, since the median, maximum, and minimum values of its boxplot are closer to the ground truth values (based on the point-to-point range measurements).

Moreover, Fig. 12 displays the number of satellites N_{SAT} for the autonomous agents of the MUAV-JS system as compared to RPS-JS, demonstrating that the number of satellites for the pursuer agent does not exceed 5, compared to the RPS-JS where the number of satellites most of the time is greater than 5. This is due to the continuous jamming operation of the MUAV-JS system, in contrast to RPS-JS, where jamming is not continuous

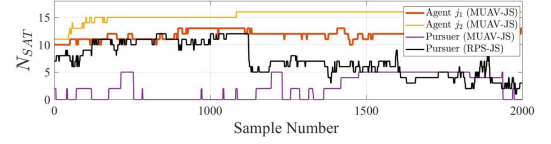


Fig. 12. N_{SAT} for pursuer drone and autonomous agents (MUAV-JS vs. RPS-JS).

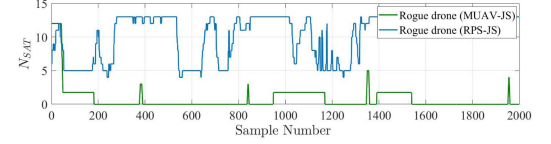


Fig. 13. N_{SAT} for rogue drone (MUAV-JS vs. RPS-JS).

TABLE I
MSE OF VARIOUS SYSTEMS UNDER JAMMING

System Used For Localization	MSE (meters)
GPS	25
IMU	21
OF	16
RPS-JS	7.7
MUAV-JS	3.5

as the SDR switches between the localization and jamming modules. Additionally, Fig. 13 depicts N_{SAT} for the rogue drone by employing either MUAV-JS or RPS-JS. It is shown that MUAV-JS is able to continuously jam the rogue drone compared to RPS-JS (i.e., for MUAV-JS N_{SAT} is always lower than 6, which is not sufficient for the drone to compute its absolute position, while for RPS-JS the rogue drone is able to extract its absolute location when the jamming procedure is disabled and the agent is utilizing the SOPs to improve self-localization).

To present the deviations in terms of localization when various systems are applied during jamming, the mean square error (MSE) for 40 outdoor experiments is tabulated in Table I. Specifically, the MSE is compared for the systems that utilize GPS, IMU, and optical flow (OF) for localization, as well as for the single- and multi-agent systems that jointly address localization and jamming, i.e., RPS-JS, and MUAV-JS, respectively, demonstrating that the MUAV-JS system is the most accurate in terms of localization (MSE equals to 3.5 m).

It should be noted that the outdoor experiments were conducted employing four UAVs (one pursuer agent, two additional system agents, and one rogue drone) in the prototype implementation, due to the unavailability of additional SDR equipment and UAVs. Nevertheless, it was important to implement the proposed system even for the specific number of agents in both hardware and software, in order to demonstrate its feasibility with real-world experiments (a short video of the experiments is also included in <http://bit.ly/3HP8xSV>). To further evaluate the performance of MUAV-JS, additional UAV agents were simulated employing the DJI drone simulator, and it is demonstrated that the proposed system's localization performance is increasing with the number of agents, as shown in Fig. 14.

VIII. CONCLUSION

This work proposes a novel online multi-agent jamming system. For localization purposes, the autonomous UAV agents are able to exploit various sensor modalities such as the SOPs, visual, optical flow, and IMU measurements, and also each agent in the system can employ the knowledge from other agents. Such

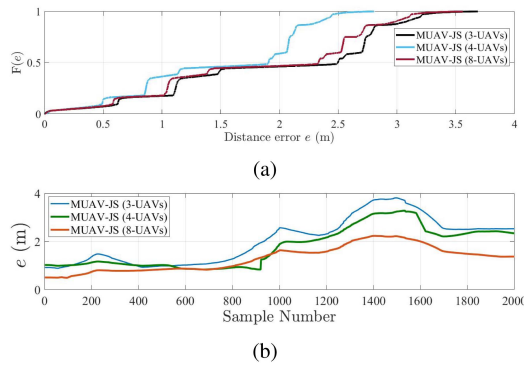


Fig. 14. (a) Position estimation error CDF and (b) Euclidean error of MUAV-JS vs. GPS+IMU using different number of agents.

a system, that allows communication and information fusion between agents, is employed to continuously detect-and-track, jam, and self-localize, aiming to counter unauthorized drone operations over critical infrastructures. An extensive evaluation of a prototype system in real outdoor scenarios showcases the system's ability to intercept rogue drones, as well as to perform online cooperative relative localization, with results that approximate the ground truth performance.

Future research includes deep learning techniques to counter issues that arise over long-term operations, such as the accumulated error from the various sensors.

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